9 Read the passage below and answer the questions that follow.

The Sun is the star at the heart of our solar system. Its gravity holds the solar system together, keeping everything – from the biggest planets to the smallest particles of debris – in its orbit. The connection and interactions between the Sun and Earth drive the seasons, ocean currents, weather, climate, radiation belts and auroras.

The Sun has produced a near-constant radiant power of 4.0×10^{26} W for around 5 billion years and is only halfway through its supply of hydrogen fuel. Hydrogen is fused into helium in the Sun's core in a three-stage process known as the proton-proton cycle.

Stage 1 Deuterium Formation**:** Two hydrogen nuclei collide to form a deuterium nucleus releasing a positron and a neutrino

Stage 2 Helium-3 Formation**:** The deuterium nucleus quickly captures another proton, creating a helium-3 nucleus and releasing a gamma-ray photon

Stage 3 Helium-4 Formation: Two helium-3 nuclei collide, resulting in one helium-4 nucleus and releasing protons back into the core

Fusion requires extremely high temperatures to give the hydrogen nuclei (protons) sufficient energy to overcome electrostatic repulsion. But even at the high temperature of 1.5×10^7 °C in the core (Figure 9.1), the average energy of the protons is not enough to allow fusion. This provided a puzzle for physicists which was finally explained by a process called quantum tunnelling.

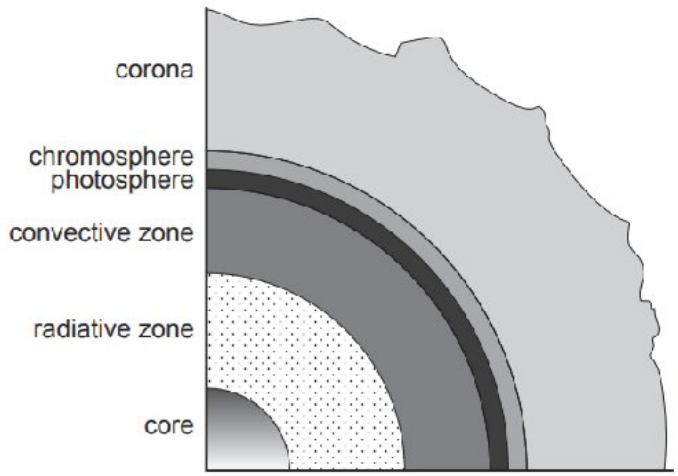


Figure 9.1 Cross section of the Sun (not to scale)

The solar wind is a continuous stream of charged particles (plasma), mainly protons and electrons, escaping the Sun's hot outer atmosphere (corona) at speeds of 500 km s^{-1}.

The Earth's magnetosphere deflects most of the solar wind (Figure 9.2). For particles that sneak through, they spiral in toward the poles, where they can contribute to the glowing lights of the auroras (Figure 9.3) when they smash into particles in the atmosphere.

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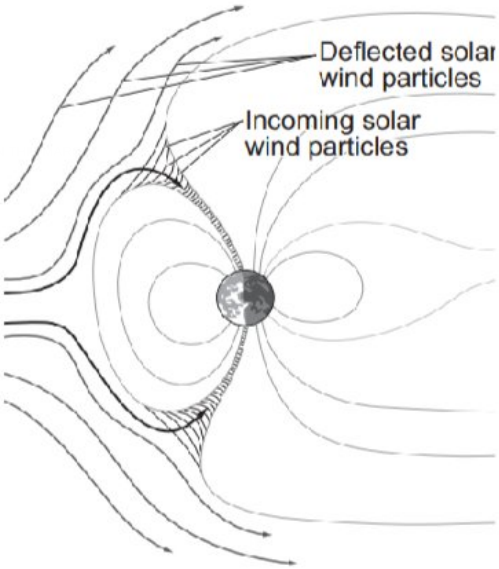


Figure 9.2 Earth's magnetic field deflecting much of the solar wind



Figure 9.3 The aurora borealis seen in Norway

Solar flares on the other hand are associated with coronal mass ejections, rapid release of plasma and ions of a typical total energy of 10^{20} to 10^{25} J. The ejected material travels through space at relativistic speeds and carries its own magnetic field which can temporarily compress Earth's magnetic field. The resulting change in Earth's magnetic field strength can burn out transformer stations in the power grid leading to blackouts. Solar research is hence an important field.

One of the most ambitious missions is the Parker Solar Probe, launched in 2018 to perform a series of passes over the Sun in a gradually changing orbit. The closest approach that the Parker Probe has made to the Sun is 1.4×10^{10} m. The power from the Sun at small distances is very large and is expected to raise the temperature of the probe to extremely high values. We can get an idea of the temperature it may reach if we make the simplifying assumption that the probe is in thermal equilibrium – that it radiates as much power per square metre as it absorbs. If this is the case, the probe would be behaving as a "black body radiator" and we can use Stefan's Law to estimate its temperature.

Stefan's Law states that the power absorbed or emitted per unit area by a black body in thermal equilibrium = σT^4, where T is the thermodynamic temperature and $\sigma = 5.7 \times 10^{-8}$ W m^{-2} K^{-4}.

(a) The equation for stage 1 of the proton-proton cycle is shown below:

$$ {}^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{2}_{1}\mathrm{H} + {}^{0}_{+1}\mathrm{e} + {}^{0}_{0}\nu $$

Write the equations for the subsequent stages:

Stage 2: .....

Stage 3: ..... [2]

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(b) (i) Assuming the protons behave like an ideal gas, show that the average kinetic energy of a proton in the core of the Sun is 3.1×10^{-16} J.

[1]

(ii) Two protons need to come within 10^{-14} m of each other to fuse.

Show that on average, the protons do not have enough energy to fuse.

Explain your reasoning clearly.

[3]

(c) (i) Explain what is meant by the wave function in quantum mechanics.

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..... [1]

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(ii) Quantum tunnelling is a quantum mechanical phenomenon in which an object such as an electron or atom passes through a potential energy barrier that, according to classical mechanics, should not be passable due to the object not having sufficient energy to pass or surmount the barrier.

Figure 9.4(a) shows a particle of energy E passing through a potential energy barrier U_0.

Figure 9.4(b) gives the wave function of the particle in the corresponding regions of space.

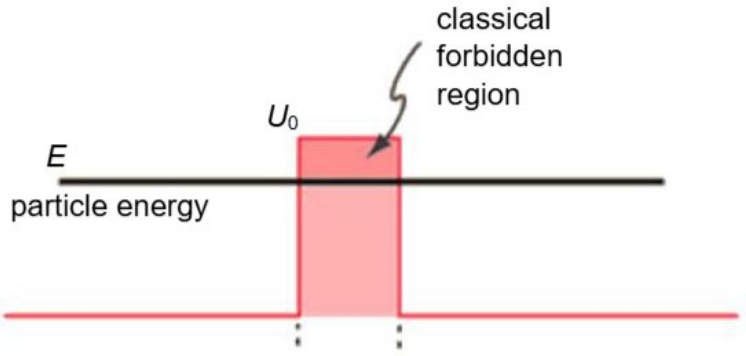


Figure 9.4(a)

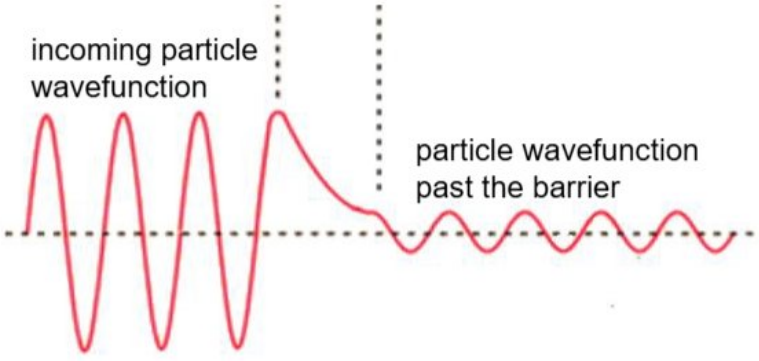


Figure 9.4(b)

With reference to Figure 9.4(b),

1. explain how the amplitude of the wave function shows the particle can exist beyond the potential barrier.

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..... [2]

2. by considering the wavelength of the wave function beyond the potential barrier, comment on the energy of the particle after quantum tunnelling.

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..... [2]

(d) Consider an electron in the solar wind travelling at an angle of 25° to the direction of the Earth's magnetic field of 40 μT as shown in Figure 9.5.

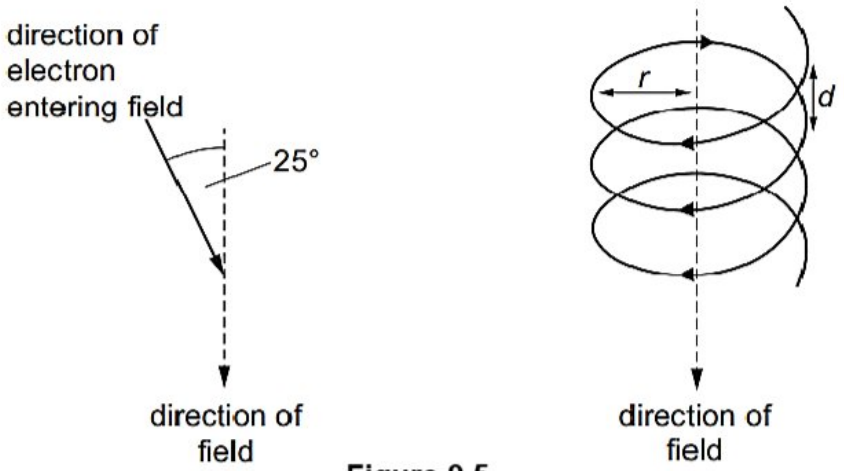


Figure 9.5

(i) By considering the horizontal and vertical components of its velocity, calculate the vertical distance *d* the electron travels between a full rotation around the field line.

distance *d* = ...m [4]

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(ii) State and explain one change to the path if it were a proton instead of an electron coming in at the same angle.

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..... [2]

(iii) Explain the mechanisms that cause the atoms in the atmosphere to produce light (aurora) when smashed by the charged particles.

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..... [2]

(e) (i) Estimate the temperature of the Parker Solar Probe at its closest distance to the Sun. Assume the Sun to be a point source of power.

temperature = ... K [3]

(ii) Suggest why missions investigating the Sun such as the Parker Solar Probe can have an importance beyond scientific discovery.

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..... [1]

[Total: 23]

~ End of Paper ~